Chương 6:
QUY TẮC ỨNG XỬ NGHỀ NGHIỆP CNTT
(Professional Code of Ethics in Information Technology)

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Tìm hiểu các tình huống sau

Kiểm soát lại

- Làm thế nào bạn quyết định hành động/quyết định trong tình huống là đúng hay sai?
- Các lý do bạn đưa ra có phù hợp với từng trường hợp không?
- Bạn đã dùng một phương pháp luận trong nhiều hay chỉ một tình huống?
- Nếu ai đó không đồng ý với bạn về câu trả lời bạn về tình huống, bạn làm thế nào để cố gắng thuyết pụ người đó rằng lập trường của bạn tạo nên có ý nghĩa như thế nào

- Đạo đức là sự phân tích hợp lý, có hệ thống về hành vi có thể gây ra lợi ích hoặc tổn hại cho người khác
- Đạo đức dựa trên l1 trí, mọi người được yêu cầu giải thích tại sao họ giữ ý kiến mà họ thực hiện
- Điều quan trọng là đạo đức đực tập trung vào những trường hợp có nhiều cách chọn lựa hoặc tự nguyện bởi vì họ có th63 quyết định/chọn lựa hành vi này thay vì một hành vi khác
- Ví dụ:
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Ví dụ:

- Bạn mua xe mới, bạn chọn màu đỏ/xanh/vàng?
- Bạn đang lái xe trên đường và bị che khuất tầm hình của một xe khác đang đậu, bạn cố gắng tránh 1 người đi bộ. Tuy nhiên sự nổ lực của bạn lại gây sự mất kiểm soát và gây ra tai nạn cho một người đi bộ khác
- Bạn bạn mất kiểm soát xe do bạn lái xe trong khi say rượu và gây tai nạn cho người khác

Các bạn hãy cho ví dụ các tình huống khác?

Thuyết tương đối

- Là lý thuyết cho rằng không có chuẩn mực đạo đức phổ quát nào đúng và sai.
  các cá nhân hoặc nhóm người khác nhau có thể hoàn toàn đối diện với một vấn đề đạo đức, và cả hai đều có thể đúng.
- là thuyết tương đối chủ quan và thuyết tương đối văn hóa. Chủ nghĩa tương đối chủ quan giữ rằng mỗi người quyết định đúng và sai do cô ta/anh ta. Khái niệm này được ghi lại trong cụm từ phổ biến, "Điều gì phù hợp với bạn có thể không phù hợp với tôi”

What are Professional Ethics?

- A set of moral (đạo đức) principles or values
- Governs (chi phối) a individual or a group
- Deals with (giao ước) what is good and bad with moral duty (trách nhiệm) and obligation (bổn phận)

What are Professional Ethics?

"Professional ethics are a code of conduct that govern how members of profession deal with each other and with third parties"

Why should we have a Professional Code of Ethics?

- A Professional Code of Ethics serves several function:
  - Symbolizes (tượng trưng cho) the professionalism (tính chuyên nghiệp nghề nghiệp) of the group
  - Defines and promotes a standard for external relations with clients and employers
  - Protects the group’s interests (lợi ích)
  - Codifies (hệ thống hóa) member’s rights
  - Expresses (diễn dạt) ideals to aspire to (khao khát)
  - Offers guidelines (đưa ra các nguyên tắc) in “gray areas”

Why have a professional code of Ethics in Computing

- Software has the potential to do good or cause harm, or to enable or influence others to do good or cause harm
- We have pride (tự hào) in our work and want the work that we do to be given recognition (công nhận) and respect (tôn trọng)
- We want to protect our livelihood (cách sinh nha)

History of the Software Engineering Code of Ethics

Who?

- The Software Engineering Code of Ethics and Professional Practice, developed by participants drawn from all the continents (lục địa) of the world, was jointly adopted by the **ACM** and the **IEEE-Computer Society** .

History of the Software Engineering Code of Ethics

Why?

- It is a standard for practicing and teaching Software Engineering and a standard for the ethical and professional development, delivery, and maintenance of software artifacts.

- The code embodies a moral commitment (cam kết đạo đức) of service to the public and is used to clarify expectations (mong đợi) and appropriate behavior of professionals.

History of the Software Engineering Code of Ethics

How?

- The code has been internationally adopted as standard of best practices by other professional computing organizations, commercial organizations, and educational institutions.

IEEE

- **I**nstitute of **E**lectrical and **E**lectronics **E**ngineers (**IEEE**)
- Viện các kỹ sư điện, điện tử quốc tế
- Tổ chức quốc tế, tập các nhà khoa học, các nhà giáo dục, các chuyên gia đầu ngành, tổ chức phi lợi nhuận
- Thành lập 1963, từ sự hợp nhất của Hiệp hội kỹ sư điện Hoa kỳ (1884) và Hiệp hội Kỹ sư Vô tuyến điện (1912)
- Hiện có 39 hội chuyên ngành với các thành viên đến từ hơn 150 nước, hoạt động trong 325 chi hội tại tất cả các vùng lãnh thổ.
- IEEE là cơ quan phát triển các tiêu chuẩn quốc tế hàng đầu với gần 900 tiêu chuẩn đã được an hành, phát hành hơn 100 tạp chí khoa học

ACM

- **A**ssociation for **C**omputing **M**achineryst
- Thành lập năm 1947
- Là một hiệp hội quốc tế về nghiên cứu, giáo dục ngành Khoa học máy tính và Tin học uy tín nhất thế giới với hơn 100.000 hội viên, tính đến năm 2011.

Eight Key Principles

Eight Key Principles

1. PUBLIC - Software engineers shall act consistently (thích hợp) with the public interest.

2. CLIENT AND EMPLOYER - Software engineers shall act in a manner (theo cách) that is in the best interests of their client and employer, consistent with the public interest.

3. PRODUCT - Software engineers shall ensure that their products and related modifications meet the highest professional standards possible.

4. JUDGMENT - Software engineers shall maintain integrity (tính liêm chính) and independence in their professional judgment.

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Eight Key Principles

5. MANAGEMENT - Software engineering managers and leaders shall subscribe to (tán thành) and promote (đẩy mạnh) an ethical approach to the management of software development and maintenance.

6. PROFESSION - Software engineers shall advance the integrity (tính liêm chính) and reputation (uy tính) of the profession consistent with the public interest.

7. COLLEAGUES - Software engineers shall be fair to and supportive of their colleagues.

8. SELF - Software engineers shall participate in lifelong learning regarding the practice of their profession and shall promote an ethical approach to the practice of the profession.

Benefits of adopting a Code of Ethics

1. Attract Employees

Attract conscientious (tận tâm) and dedicated (cống hiến) employees who want to be involved (tham gia vào) in an organization that produces quality software.

2. Public Concern

Having a reputation (uy tính/danh tiếng) of quality and dependability (tính tin cậy) will promote (đẩy mạnh) an ethical (đúng theo nguyên tắc ứng xử) image for your company. In return, this will let the public know that the corporation (các hiệp hội) is working for the public good (lợi ích) and proudly accepts (tự hào nhận) that responsibility.

Benefits of adopting a Code of Ethics

3. Professional Image (hình ảnh nghề nghiệp)

Conforming (tuân thủ) to quality standards (tiêu chuẩn chất lượng) will gain respectability (nhận được sự tôn trọng) for your corporation and improve the quality of software that it produces.

4. Public Trust (lòng tin của công chúng)

Producing a quality software will inform the public that their best interests are being met with confidentiality and high standards.

5. Internal Standards

Improve communications between management and their colleagues. Software engineers shall make these practices a lifelong practice and produce quality products that will reflect in every aspect of their professional life.

Steps in Adopting the Code

- Carefully read the Code of Ethics.
- Verify that your organization is willing (sẳn long) to follow the Code of Ethics.
- Complete and submit the online application .
- Incorporate (hợp nhất) the Code of Ethics into your organization.
- Feel free to include the SECEPP logo in your organization's documents and post the mini-poster at your location to demonstrate your organization's dedication to excellence.

1. Public

PUBLIC –Software engineers shall act consistently with the public interest.

1.01. Accept full responsibility for their own work.

1.02. Moderate the interests of the software engineer, the employer, the client and the users with the public good.

1.03. Approve software only if they have a well-founded belief that it is safe, meets specifications, passes appropriate tests, and does not diminish quality of life, diminish privacy or harm the environment. The ultimate effect of the work should be to the public good.

1.04. Disclose to appropriate persons or authorities any actual or potential danger to the user, the public, or the environment, that they reasonably believe to be associated with software or related documents.

Principle 1: PUBLIC

PUBLIC –Software engineers shall act consistently with the public interest.

1.05. Cooperate in efforts to address matters of grave public concern caused by software, its installation, maintenance, support or documentation.

1.06. Be fair and avoid deception in all statements, particularly public ones, concerning software or related documents, methods and tools.

1.07. Consider issues of physical disabilities, allocation of resources, economic disadvantage and other factors that can diminish access to the benefits of software.

1.08. Be encouraged to volunteer professional skills to good causes and to contribute to public education concerning the discipline.

Principle 2: CLIENT AND EMPLOYER

CLIENT AND EMPLOYER - Software engineers shall act in a manner (theo cách) that is in the best interests of their client and employer, consistent with the public interest.

2.01. Provide service in their areas of competence, being honest (trung thực) and forthright (thẳng thắng) about any limitations of their experience and education.

2.02. Not knowingly use software that is obtained or retained either illegally or unethically.

2.03. Use the property of a client or employer only in ways properly authorized, and with the client's or employer's knowledge and consent.

2.04. Ensure that any document upon which they rely has been approved, when required, by someone authorized to approve it.

Principle 2: CLIENT AND EMPLOYER

2.05. Keep private any confidential information gained in their professional work, where such confidentiality is consistent with the public interest and consistent with the law.

2.06. Identify, document, collect evidence and report to the client or the employer promptly if, in their opinion**, a project is likely to fail, to prove too expensive, to violate intellectual property law, or otherwise to be problematic.**

2.07. Identify, document, and report significant issues of social concern, of which they are aware, in software or related documents, to the employer or the client.

Principle 2: CLIENT AND EMPLOYER

2.08. Accept no outside work detrimental to the work they perform for their primary employer.

2.09. Promote no interest adverse to their employer or client, unless a higher ethical concern is being compromised; in that case, inform the employer or another appropriate authority of the ethical concern.

Principle 3: PRODUCT

PRODUCT - Software engineers shall ensure that their products and related modifications meet the highest professional standards (tiêu chuẩn nghề nghiệp) possible

3.01. Strive for high quality, acceptable cost, and a reasonable schedule, ensuring significant tradeoffs are clear to and accepted by the employer and the client, and are available for consideration by the user and the public.

3.02. Ensure proper and achievable goals and objectives for any project on which they work or propose.

3.03. Identify, define and address ethical, economic, cultural, legal and environmental issues related to work projects.

Principle 3: PRODUCT

3.04. Ensure that they are qualified for any project on which they work or propose to work, by an appropriate combination of education, training, and experience.

3.05. Ensure that an appropriate method is used for any project on which they work or propose to work.

3.06. Work to follow professional standards, when available, that are most appropriate for the task at hand, departing from these only when ethically or technically justified.

3.07. Strive to fully understand the specifications for software on which they work.

Principle 3: PRODUCT

3.08. Ensure that specifications for software on which they work have been well documented, satisfy the users' requirements and have the appropriate approvals.

3.09. Ensure realistic quantitative estimates of cost, scheduling, personnel, quality and outcomes on any project on which they work or propose to work and provide an uncertainty assessment of these estimates.

3.10. Ensure adequate testing, debugging, and review of software and related documents on which they work.

3.11. Ensure adequate documentation, including significant problems discovered and solutions adopted, for any project on which they work.

Principle 3: PRODUCT

3.12. Work to develop software and related documents that respect the privacy of those who will be affected by that software.

3.13. Be careful to use only accurate data (dữ liệu đúng) derived by ethical and lawful means (phương tiện hợp pháp và đạo đức), and use it only in ways properly authorized (đúng thẩm quyền).

3.14. Maintain the integrity of data, being sensitive to outdated or flawed occurrences.

3.15 Treat all forms of software maintenance with the same professionalism as new development.

Principle 4: JUDGMENT

JUDGMENT - Software engineers shall maintain integrity and independence in their professional judgment.

4.01. Temper all technical judgments by the need to support and maintain human values.

4.02 Only endorse documents either prepared under their supervision or within their areas of competence and with which they are in agreement.

4.03. Maintain professional objectivity with respect to any software or related documents they are asked to evaluate.

4.04. Not engage in deceptive financial practices such as bribery, double billing, or other improper financial practices.

4.05. Disclose to all concerned parties those conflicts of interest that cannot reasonably be avoided or escaped.

Principle 4: JUDGMENT

JUDGMENT - Software engineers shall maintain integrity and independence in their professional judgment.

4.06. Refuse to participate, as members or advisors, in a private, governmental or professional body concerned with software related issues, in which they, their employers or their clients have undisclosed potential conflicts of interest

Principle 5: MANAGEMENT

MANAGEMENT - Software engineering managers and leaders shall subscribe to and promote an ethical approach to the management of software development and maintenance.

5.01 Ensure good management for any project on which they work, including effective procedures for promotion of quality and reduction of risk.

5.02. Ensure that software engineers are informed of standards before being held to them.

5.03. Ensure that software engineers know the employer's policies and procedures for protecting passwords, files and information that is confidential to the employer or confidential to others.

Principle 5: MANAGEMENT

5.04. Assign work only after taking into account appropriate contributions of education and experience tempered with a desire to further that education and experience.

5.05. Ensure realistic quantitative estimates of cost, scheduling, personnel, quality and outcomes on any project on which they work or propose to work, and provide an uncertainty assessment of these estimates.

5.06. Attract potential software engineers only by full and accurate description of the conditions of employment.

5.07. Offer fair and just remuneration.

5.08. Not unjustly prevent someone from taking a position for which that person is suitably qualified.

Principle 5: MANAGEMENT

5.09. Ensure that there is a fair agreement concerning ownership of any software, processes, research, writing, or other intellectual property to which a software engineer has contributed.

5.10. Provide for due process in hearing charges of violation of an employer's policy or of this Code.

5.11. Not ask a software engineer to do anything inconsistent with this Code.

5.12. Not punish anyone for expressing ethical concerns about a project.

Principle 6: PROFESSION

PROFESSION - Software engineers shall advance the integrity (tính liêm chính) and reputation (uy tính) of the profession consistent with the public interest.

6.01. Help develop an organizational environment favorable to acting ethically.

6.02. Promote public knowledge of software engineering.

6.03. Extend software engineering knowledge by appropriate participation in professional organizations, meetings and publications.

6.04. Support, as members of a profession, other software engineers striving to follow this Code.

6.05. Not promote their own interest at the expense of the profession, client or employer.

Principle 6: PROFESSION

6.06. Obey all laws governing their work, unless, in exceptional circumstances, such compliance is inconsistent with the public interest.

6.07. Be accurate in stating the characteristics of software on which they work, avoiding not only false claims but also claims that might reasonably be supposed to be speculative, vacuous, deceptive, misleading, or doubtful.

6.08. Take responsibility for detecting, correcting, and reporting errors in software and associated documents on which they work.

6.09. Ensure that clients, employers, and supervisors know of the software engineer's commitment to this Code of ethics, and the subsequent ramifications of such commitment.

Principle 6: PROFESSION

6.10. Avoid associations with businesses and organizations which are in conflict with this code.

6.11. Recognize that violations of this Code are inconsistent with being a professional software engineer.

6.12. Express concerns to the people involved when significant violations of this Code are detected unless this is impossible, counter-productive, or dangerous.

6.13. Report significant violations of this Code to appropriate authorities when it is clear that consultation with people involved in these significant violations is impossible, counter-productive or dangerous.

Principle 7: COLLEAGUES

COLLEAGUES - Software engineers shall be fair to and supportive of their colleagues

7.01. Encourage colleagues to adhere to this Code.

7.02. Assist colleagues in professional development.

7.03. Credit fully the work of others and refrain from taking undue credit.

7.04. Review the work of others in an objective, candid, and properly-documented way.

7.05. Give a fair hearing to the opinions, concerns, or complaints of a colleague.

Principle 7: COLLEAGUES

7.06. Assist colleagues in being fully aware of current standard work practices including policies and procedures for protecting passwords, files and other confidential information, and security measures in general.

7.07. Not unfairly intervene in the career of any colleague; however, concern for the employer, the client or public interest may compel software engineers, in good faith, to question the competence of a colleague.

7.08. In situations outside of their own areas of competence, call upon the opinions of other professionals who have competence in that area.

Principle 8: SELF

SELF - Software engineers shall participate in lifelong learning regarding the practice of their profession and shall promote an ethical approach to the practice of the profession.

8.01. Further their knowledge of developments in the analysis, specification, design, development, maintenance and testing of software and related documents, together with the management of the development process.

8.02. Improve their ability to create safe, reliable, and useful quality software at reasonable cost and within a reasonable time.

8.03. Improve their ability to produce accurate, informative, and well-written documentation.

Principle 8: SELF

8.04. Improve their understanding of the software and related documents on which they work and of the environment in which they will be used.

8.05. Improve their knowledge of relevant standards and the law governing the software and related documents on which they work.

8.06 Improve their knowledge of this Code, its interpretation, and its application to their work.

8.07 Not give unfair treatment to anyone because of any irrelevant prejudices.

8.08. Not influence others to undertake any action that involves a breach of this Code.

8.09. Recognize that personal violations of this Code are inconsistent with being a professional software engineer.